

Exercise: LangChain Agents & Tools

1. Objective

To build an AI assistant that can independently look up company policies, check order statuses in a database, and provide a seamless response to the customer.

2. Building the Knowledge Base from Local Text

The Problem: Support staff spend 40% of their time answering the same questions about shipping and returns.

- **The Goal:** Give the AI "knowledge" by feeding it a simple text source.
- **Action:** You will create a plain text block within your code (or a simple .txt file) containing your store's specific rules. You will then set up a **Retriever**—a system that acts like a librarian, scanning your text to find the exact sentence needed to answer a customer's question. This allows the AI to stay grounded in facts rather than making up policies.

3. Connecting to Internal Systems

The Problem: Customers frequently ask "Where is my order?" which requires checking a private database.

- **The Goal:** Give the AI a specific "tool" to look up order IDs.
- **Action:** You will create a custom function that simulates a database lookup. You must describe this tool clearly (using a docstring) so the AI knows it should use it only when it sees an Order ID.

4. Setting up the "Brain" (The Agent)

The Problem: A standard chatbot just talks; it doesn't "act."

- **The Goal:** Build an Agent that can decide *which* tool to use based on the customer's intent.
- **Action:** You will configure the AI's persona. It needs to be polite, concise, and smart enough to realize when it needs to look at your provided text vs. when it needs to check an Order ID.

5. Monitoring the Process (The Middleware)

The Problem: In a real business, we need to see what the AI is "thinking" to ensure it isn't making mistakes.

- **The Goal:** Create a logging window that shows the AI's internal logic.
- **Action:** You will implement a "Middleware" component. This will print a message every time the AI starts a thought process and every time it finishes a task, allowing you to see the "gears turning" behind the scenes before the customer gets the final answer.

6. Live Simulation

The Goal: Test the assistant with real-world scenarios.

- **Test 1:** Ask about shipping costs.
- **Test 2:** Ask for the status of a specific order number.
- **Test 3:** Combine them: "Check order 101 and tell me if I can still return it."

7. Scaling to Enterprise

The Problem: Your company is growing. Policies are no longer just a few lines of text; they are now stored in long files in a **Google Drive** folder. Furthermore, management wants a **follow-up email** sent to the customer automatically after every resolved query.

Your Task:

1. **Research the Drive Connection:** How would you change your "Document Loader" to pull from a Google Drive folder instead of a local text string?
2. **Longer Content:** Change the implementation to store the information in Vector database, use text splitters
3. **The Email Trigger:** Take the AI's final answer and send it to a customer's email address.